# AiProfessor MVP - Entity Relationship Diagram

PostgreSQL + pgvector proposal | 16 tables | Source: MVP table specification v0.1 | 2026-11-02

A1 landscape - vector PDF

### users — IDENTITY

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| UK | email | VARCHAR(320) | N |
| - | password_hash | VARCHAR(255) | N |
| - | display_name | VARCHAR(50) | N |
| - | role | VARCHAR(20) | N |
| - | status | VARCHAR(20) | N |
| - | last_login_at | TIMESTAMPTZ | Y |
| - | withdrawn_at | TIMESTAMPTZ | Y |
| - | created_at | TIMESTAMPTZ | N |
| - | updated_at | TIMESTAMPTZ | N |

### refresh_tokens — IDENTITY

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | user_id | UUID | N |
| UK | token_hash | VARCHAR(255) | N |
| - | token_family_id | UUID | N |
| - | expires_at | TIMESTAMPTZ | N |
| - | revoked_at | TIMESTAMPTZ | Y |
| FK | replaced_by_token_id | UUID | Y |
| - | created_ip_hash | VARCHAR(128) | Y |
| - | user_agent | VARCHAR(500) | Y |
| - | created_at | TIMESTAMPTZ | N |

### documents — DOCUMENT

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | user_id | UUID | N |
| - | original_file_name | VARCHAR(255) | N |
| UK | storage_key | VARCHAR(500) | N |
| - | mime_type | VARCHAR(100) | N |
| - | file_size_bytes | BIGINT | N |
| - | page_count | INTEGER | Y |
| - | sha256 | CHAR(64) | N |
| - | status | VARCHAR(30) | N |
| - | failure_code | VARCHAR(100) | Y |
| - | failure_message | VARCHAR(500) | Y |
| - | processed_at | TIMESTAMPTZ | Y |
| - | deleted_at | TIMESTAMPTZ | Y |
| - | created_at | TIMESTAMPTZ | N |
| - | updated_at | TIMESTAMPTZ | N |

### ai_usages — AI

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | user_id | UUID | Y |
| FK | document_id | UUID | Y |
| FK | job_id | UUID | Y |
| - | operation | VARCHAR(30) | N |
| - | provider | VARCHAR(50) | N |
| - | model | VARCHAR(100) | N |
| - | prompt_version | VARCHAR(50) | Y |
| - | input_tokens | INTEGER | N |
| - | output_tokens | INTEGER | N |
| - | latency_ms | INTEGER | N |
| - | estimated_cost | NUMERIC(18,6) | Y |
| - | currency | CHAR(3) | Y |
| - | success | BOOLEAN | N |
| - | error_code | VARCHAR(100) | Y |
| - | request_id | VARCHAR(255) | Y |
| - | created_at | TIMESTAMPTZ | N |

### processing_jobs — JOB

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | user_id | UUID | N |
| FK | document_id | UUID | Y |
| - | job_type | VARCHAR(30) | N |
| - | target_type | VARCHAR(30) | N |
| - | target_id | UUID | N |
| UK | idempotency_key | VARCHAR(100) | N |
| - | status | VARCHAR(20) | N |
| - | current_step | VARCHAR(50) | Y |
| - | progress_percent | SMALLINT | N |
| - | retry_count | SMALLINT | N |
| - | max_retries | SMALLINT | N |
| - | error_code | VARCHAR(100) | Y |
| - | error_message | VARCHAR(500) | Y |
| - | next_retry_at | TIMESTAMPTZ | Y |
| - | started_at | TIMESTAMPTZ | Y |
| - | completed_at | TIMESTAMPTZ | Y |
| - | created_at | TIMESTAMPTZ | N |
| - | updated_at | TIMESTAMPTZ | N |

### processing_job_attempts — JOB

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | job_id | UUID | N |
| UK | attempt_no | SMALLINT | N |
| - | worker_id | VARCHAR(100) | Y |
| - | status | VARCHAR(20) | N |
| - | error_code | VARCHAR(100) | Y |
| - | error_message | VARCHAR(500) | Y |
| - | started_at | TIMESTAMPTZ | N |
| - | finished_at | TIMESTAMPTZ | Y |

### document_chunks — RAG

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | document_id | UUID | N |
| UK | chunk_index | INTEGER | N |
| - | page_start | INTEGER | N |
| - | page_end | INTEGER | N |
| - | content | TEXT | N |
| - | token_count | INTEGER | Y |
| - | content_hash | CHAR(64) | N |
| - | chunk_strategy | VARCHAR(50) | N |
| - | chunk_size | INTEGER | N |
| - | chunk_overlap | INTEGER | N |
| - | created_at | TIMESTAMPTZ | N |

### chunk_embeddings — RAG

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | chunk_id | UUID | N |
| - | provider | VARCHAR(50) | N |
| - | model | VARCHAR(100) | N |
| UK | model_version | VARCHAR(100) | N |
| - | dimensions | INTEGER | N |
| - | embedding | VECTOR(n) | Y |
| UK | external_vector_id | VARCHAR(255) | Y |
| - | created_at | TIMESTAMPTZ | N |

### summaries — SUMMARY

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | document_id | UUID | N |
| FK | user_id | UUID | N |
| FK | job_id | UUID | Y |
| - | summary_type | VARCHAR(20) | N |
| - | topic | VARCHAR(255) | Y |
| - | title | VARCHAR(255) | N |
| - | content | TEXT | Y |
| - | status | VARCHAR(20) | N |
| - | insufficient_evidence | BOOLEAN | N |
| - | provider | VARCHAR(50) | Y |
| - | model | VARCHAR(100) | Y |
| - | prompt_version | VARCHAR(50) | Y |
| - | generation_config | JSONB | Y |
| - | failure_code | VARCHAR(100) | Y |
| - | created_at | TIMESTAMPTZ | N |
| - | updated_at | TIMESTAMPTZ | N |

### summary_citations — SUMMARY

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | summary_id | UUID | N |
| FK | chunk_id | UUID | N |
| UK | citation_order | INTEGER | N |
| - | claim_text | TEXT | Y |
| - | quote_text | TEXT | N |
| - | page_start | INTEGER | N |
| - | page_end | INTEGER | N |
| - | created_at | TIMESTAMPTZ | N |

### quiz_sets — QUIZ

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | document_id | UUID | N |
| FK | user_id | UUID | N |
| FK | job_id | UUID | Y |
| - | topic | VARCHAR(255) | Y |
| - | difficulty | VARCHAR(20) | N |
| - | mcq_count | SMALLINT | N |
| - | saq_count | SMALLINT | N |
| - | status | VARCHAR(20) | N |
| - | provider | VARCHAR(50) | Y |
| - | model | VARCHAR(100) | Y |
| - | prompt_version | VARCHAR(50) | Y |
| - | failure_code | VARCHAR(100) | Y |
| - | created_at | TIMESTAMPTZ | N |
| - | updated_at | TIMESTAMPTZ | N |

### questions — QUIZ

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | quiz_set_id | UUID | N |
| FK | document_id | UUID | N |
| UK | question_order | SMALLINT | N |
| - | question_type | VARCHAR(10) | N |
| - | difficulty | VARCHAR(20) | N |
| - | topic | VARCHAR(255) | Y |
| - | question_text | TEXT | N |
| - | reference_answer | TEXT | N |
| - | explanation | TEXT | N |
| - | grading_rubric | JSONB | Y |
| - | provider | VARCHAR(50) | N |
| - | model | VARCHAR(100) | N |
| - | prompt_version | VARCHAR(50) | N |
| - | created_at | TIMESTAMPTZ | N |

### question_choices — QUIZ

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | question_id | UUID | N |
| UK | choice_order | SMALLINT | N |
| - | content | TEXT | N |
| - | is_correct | BOOLEAN | N |
| - | created_at | TIMESTAMPTZ | N |

### question_citations — QUIZ

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | question_id | UUID | N |
| FK | chunk_id | UUID | N |
| UK | citation_order | INTEGER | N |
| - | quote_text | TEXT | N |
| - | page_start | INTEGER | N |
| - | page_end | INTEGER | N |
| - | created_at | TIMESTAMPTZ | N |

### question_attempts — LEARNING

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | question_id | UUID | N |
| FK | user_id | UUID | N |
| UK | attempt_no | INTEGER | N |
| FK | selected_choice_id | UUID | Y |
| - | submitted_text | TEXT | Y |
| - | grading_status | VARCHAR(20) | N |
| - | is_correct | BOOLEAN | Y |
| - | score | NUMERIC(5,2) | Y |
| - | feedback | TEXT | Y |
| - | missing_keywords | JSONB | Y |
| - | provider | VARCHAR(50) | Y |
| - | model | VARCHAR(100) | Y |
| - | prompt_version | VARCHAR(50) | Y |
| - | submitted_at | TIMESTAMPTZ | N |
| - | graded_at | TIMESTAMPTZ | Y |

### review_items — LEARNING

| | | | |
|---|---|---|---|
| PK | id | UUID | N |
| FK | user_id | UUID | N |
| FK | question_id | UUID | N |
| - | reason | VARCHAR(20) | N |
| - | status | VARCHAR(20) | N |
| FK | latest_attempt_id | UUID | Y |
| - | retry_count | INTEGER | N |
| - | last_reviewed_at | TIMESTAMPTZ | Y |
| - | created_at | TIMESTAMPTZ | N |
| - | updated_at | TIMESTAMPTZ | N |

target_type + target_id

## RELATIONSHIP REFERENCE

All entries are 1:N unless marked otherwise. Child column is shown after the arrow.

### IDENTITY & OWNERSHIP

- users -> refresh_tokens.user_id
- refresh_tokens -> refresh_tokens.replaced_by_token_id (0..1)
- users -> documents.user_id
- users -> processing_jobs.user_id
- users -> summaries.user_id
- users -> quiz_sets.user_id
- users -> question_attempts.user_id
- users -> review_items.user_id
- users -> ai_usages.user_id (optional)

### DOCUMENT, JOB & RAG

- documents -> processing_jobs.document_id (optional)
- processing_jobs -> processing_job_attempts.job_id
- documents -> document_chunks.document_id
- document_chunks -> chunk_embeddings.chunk_id
- documents -> ai_usages.document_id (optional)
- processing_jobs -> ai_usages.job_id (optional)
- processing_jobs -> target_type + target_id (logical)

### SUMMARY & QUIZ GENERATION

- documents -> summaries.document_id
- processing_jobs -> summaries.job_id (optional)
- summaries -> summary_citations.summary_id
- document_chunks -> summary_citations.chunk_id
- documents -> quiz_sets.document_id
- processing_jobs -> quiz_sets.job_id (optional)
- quiz_sets -> questions.quiz_set_id
- documents -> questions.document_id

### LEARNING & CITATIONS

- questions -> question_choices.question_id
- questions -> question_citations.question_id
- document_chunks -> question_citations.chunk_id
- questions -> question_attempts.question_id
- question_choices -> question_attempts.selected_choice_id (optional)
- questions -> review_items.question_id (0..1 per user)
- question_attempts -> review_items.latest_attempt_id (optional)

## LEGEND

PK primary key | FK foreign key | UK unique key / unique component | physical FK, 1:N | polymorphic reference

Null column: N = NOT NULL, Y = nullable | Composite UK components are highlighted individually.

Design note: actual FK constraints, ON DELETE actions, vector dimensions, and composite unique constraints must be finalized in DDL/ADR.